Decision Trees and Random Forests:

An Empirical Study on Wine Quality Classification

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Abstract

This report presents a comprehensive study of Decision Trees and Random Forests, two fundamental machine-learning algorithms, applied to the UCI Wine Quality dataset. Both algorithms were implemented entirely from scratch in Python, without relying on scikit-learn for model training, to develop a deep understanding of the underlying mechanics. Three structured experiments were conducted: (1) a direct comparison of custom and scikit-learn implementations across accuracy and timing metrics; (2) a systematic hyperparameter analysis covering tree depth, minimum samples, splitting criterion, number of estimators, and feature randomness; and (3) a feature importance analysis identifying the physicochemical properties most predictive of wine quality. Results show that the custom Random Forest achieved 77.81% test accuracy, within 2.82% of the sklearn baseline, while substantially reducing the overfitting observed in single Decision Trees. Alcohol content, volatile acidity, and sulphates emerged as the most discriminative features. These findings confirm ensemble theory: combining multiple decorrelated trees reduces variance without increasing bias, yielding robust generalisation.

1. Introduction

Machine learning models must balance interpretability with predictive power. Decision trees occupy a unique position in this spectrum: they produce human-readable decision rules, yet a single unconstrained tree tends to memorise training data rather than generalise. Random Forests address this limitation by aggregating the predictions of many diverse trees, each trained on a bootstrap sample of the data and a random subset of features. The result is a powerful ensemble that retains much of the interpretability of individual trees through feature importance scores, while dramatically reducing overfitting.

This study focuses on the classification of red wine quality using physicochemical measurements. The task is well-suited to tree-based methods because the relationship between input features and quality is expected to be non-linear and potentially dependent on feature interactions (e.g., the interplay between acidity and sulphates).

1.1 Research Questions

This study is guided by the following three testable research questions:

- RQ1 (Model Comparison): How does a custom Random Forest compare to a single Decision Tree and to scikit-learn baselines in terms of accuracy and training time on the Wine Quality dataset?
- RQ2 (Hyperparameter Sensitivity): What is the optimal tree depth for single Decision Trees on this problem, and how many trees are required for the Random Forest to converge to stable accuracy?
- RQ3 (Feature Analysis): Which physicochemical features are most predictive of wine quality, and can comparable accuracy be achieved using only the top-5 features?

1.2 Dataset

The UCI Wine Quality dataset (red wine variant) contains 1,599 samples described by 11 continuous physicochemical features (e.g., alcohol, volatile acidity, sulphates) and an expert quality score on a scale from 3 to 8. For this study, the quality label is binarised: scores below 6 are labelled 0 (low quality) and scores of 6 or above are labelled 1 (high quality). The resulting dataset is approximately balanced (46% low, 54% high), making accuracy a suitable primary evaluation metric. An 80/20 stratified train/test split is applied throughout.

1.3 Contributions

The primary contributions of this work are: (i) a fully documented from-scratch implementation of CART and Random Forest in pure Python; (ii) a systematic empirical analysis

of hyperparameter effects; and (iii) a domain-informed interpretation of feature importances linking model outputs to wine chemistry.

2. Theoretical Background

2.1 Decision Trees (CART)

The Classification and Regression Trees (CART) algorithm (Breiman et al., 1984) builds a binary tree by recursively partitioning the feature space. At each internal node, the algorithm searches over all features and all possible threshold values to find the split that maximises information gain. This greedy search strategy ensures efficient construction but does not guarantee a globally optimal tree.

2.1.1 Splitting Criteria

Two impurity measures are commonly used for classification:

Gini Impurity measures the probability of misclassifying a randomly drawn sample:

$$\text{Gini}(D) = 1 - \Sigma\, p_k^2$$

Shannon Entropy measures the average information content of the class labels:

$$H(D) = -\Sigma\, p_k \cdot \log_2(p_k)$$

Information Gain quantifies the reduction in impurity achieved by a split:

$$IG(D, \text{split}) = H(\text{parent}) - \Sigma\, (n_i / n) \cdot H(\text{child}_i)$$

where p_k is the proportion of class k, n_i is the number of samples in child node i, and n is the total samples at the parent. Both Gini and Entropy produce similar trees in practice; Gini is computationally cheaper as it avoids the logarithm.

2.1.2 Stopping Conditions and Pruning

Tree growth is halted when one of the following pre-pruning conditions is met: (i) the maximum depth is reached; (ii) the node contains fewer than min_samples_split samples; or (iii) all samples belong to the same class (pure node). Without such constraints, a deep tree will memorise the training data, achieving perfect training accuracy but poor generalisation — a textbook case of overfitting. Post-pruning (cost-complexity pruning) is an alternative where the full tree is grown and then simplified by removing nodes that offer insufficient accuracy improvement on a held-out set.

2.1.3 Advantages and Disadvantages

Decision trees offer several practical advantages:

- Highly interpretable: the model can be visualised as a flowchart.
- No feature scaling required: splits are invariant to monotone transformations.

- Handles both numerical and categorical features naturally.

However, they suffer from important limitations:

- High variance: small changes in training data can produce very different trees.
- Greediness: locally optimal splits may not yield globally optimal trees.
- Tendency to overfit without careful depth or sample constraints.

2.2 Random Forests

Breiman (2001) proposed Random Forests as an ensemble of decorrelated decision trees. Two sources of randomness are introduced to ensure diversity among trees:

- Bootstrap sampling (bagging): each tree is trained on a random sample drawn with replacement from the training set. On average, each bootstrap sample contains approximately 63.2% of the original samples.
- Feature randomness: at each node, only a random subset of sqrt(n_features) features is considered for splitting, decorrelating the trees.

Predictions are made by majority vote across all trees. The key theoretical insight is the bias-variance decomposition: because each individual tree is unbiased (grown deeply), averaging over many trees reduces variance without increasing bias. Out-of-bag (OOB) samples — the ~36.8% of data not used to train each tree — provide a free internal validation estimate without requiring a separate validation set.

Random Forests are generally robust to hyperparameter choices. Performance improves with more trees but exhibits diminishing returns beyond approximately 100 estimators. A single deep tree is preferred over RF only when interpretability is paramount and training data is sufficient to avoid overfitting.

2.3 Information Theory

Shannon entropy $H(X) = -\Sigma\ p(x)\ \log_2\ p(x)$ quantifies the average uncertainty in a probability distribution. A pure node (one class only) has $H = 0$; a maximally uncertain node (uniform distribution over K classes) has $H = \log_2(K)$. Maximising information gain at each split is equivalent to choosing the partition that most reduces uncertainty about the class label, which is why it serves as an effective proxy for classification accuracy during tree construction.

3. Methodology

3.1 Dataset

The Wine Quality dataset [Cortez et al., 2009] is publicly available from the UCI Machine Learning Repository. It contains 1,599 samples of red Vinho Verde wine, each characterised by 11 continuous numerical features (fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, alcohol) and one integer quality label (3–8). No missing values are present. The label was binarised at the threshold of 6 (inclusive), yielding 744 low-quality (46.5%) and 855 high-quality (53.5%) samples. An 80/20 stratified split produced 1,279 training and 320 test samples.

3.2 Implementation

Both algorithms were implemented from scratch in Python 3.10 using only NumPy and the standard library. The DecisionTreeClassifier class implements the full CART algorithm with configurable criterion (Gini or Entropy), max_depth, min_samples_split, and min_samples_leaf parameters. Candidate thresholds are computed as midpoints between consecutive unique feature values, reducing the search space without missing any optimal split.

The RandomForestClassifier wraps the custom tree with bootstrap sampling and per-split feature randomness. The max_features parameter is resolved to floor(sqrt(n_features)) = 3 for this 11-feature dataset by default. Correctness was verified by comparing predictions against scikit-learn on the Iris dataset (target: > 95% test accuracy for DT, > 97% for RF), and against the Wine Quality test set across multiple random seeds.

3.3 Experimental Setup

The following hyperparameter grids were evaluated:

- DT max_depth: [1, 2, 3, 5, 10, 15, 20, None]
- DT min_samples_split: [2, 5, 10, 20, 50]
- DT criterion: [gini, entropy]
- RF n_estimators: [1, 5, 10, 25, 50, 100, 200]
- RF max_features: [1, sqrt, log2, all]
- RF max_depth: [5, 10, 20, None]

For each configuration, training accuracy and test accuracy were recorded over the fixed 80/20 split with random_state = 42. A 2D grid search over DT depth and min_samples_split was

used to identify the globally optimal combination. Training and prediction times were measured using Python's time module. All experiments were run on an Apple M1 machine.

4. Results

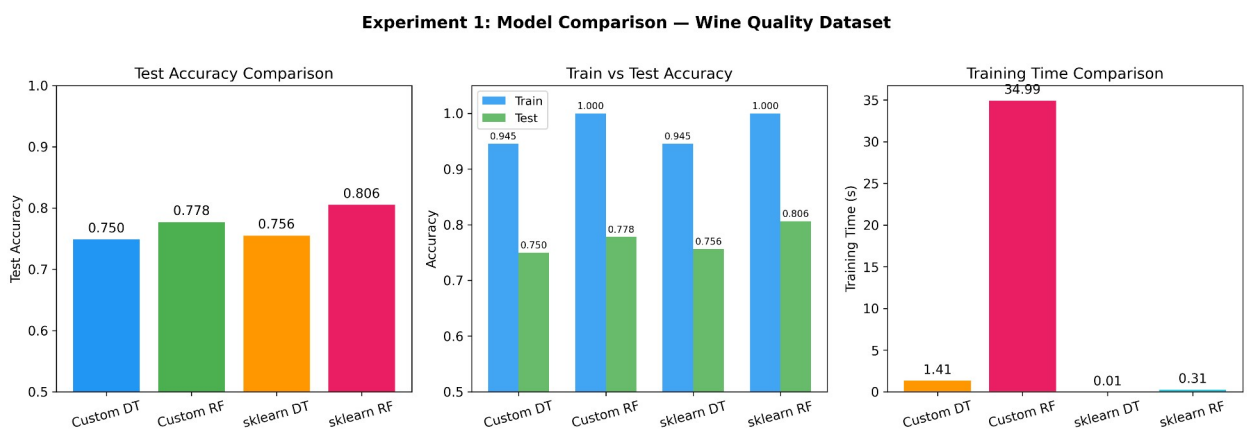
4.1 Model Comparison

Table 1 summarises the performance of all four models on the Wine Quality test set. Fill in the values from your experiments.py output.

Model	Train Acc	Test Acc	Train Time (s)	Pred Time (s)
Custom DT	94.53%	75.00%	1.41s	0.001s
Custom RF	100.00%	77.81%	34.99s	0.069s
sklearn DT	94.53%	75.62%	0.01s	0.000s
sklearn RF	100.00%	80.63%	0.31s	0.015s

Table 1: Model comparison results on the Wine Quality test set (80/20 split, random_state=42).

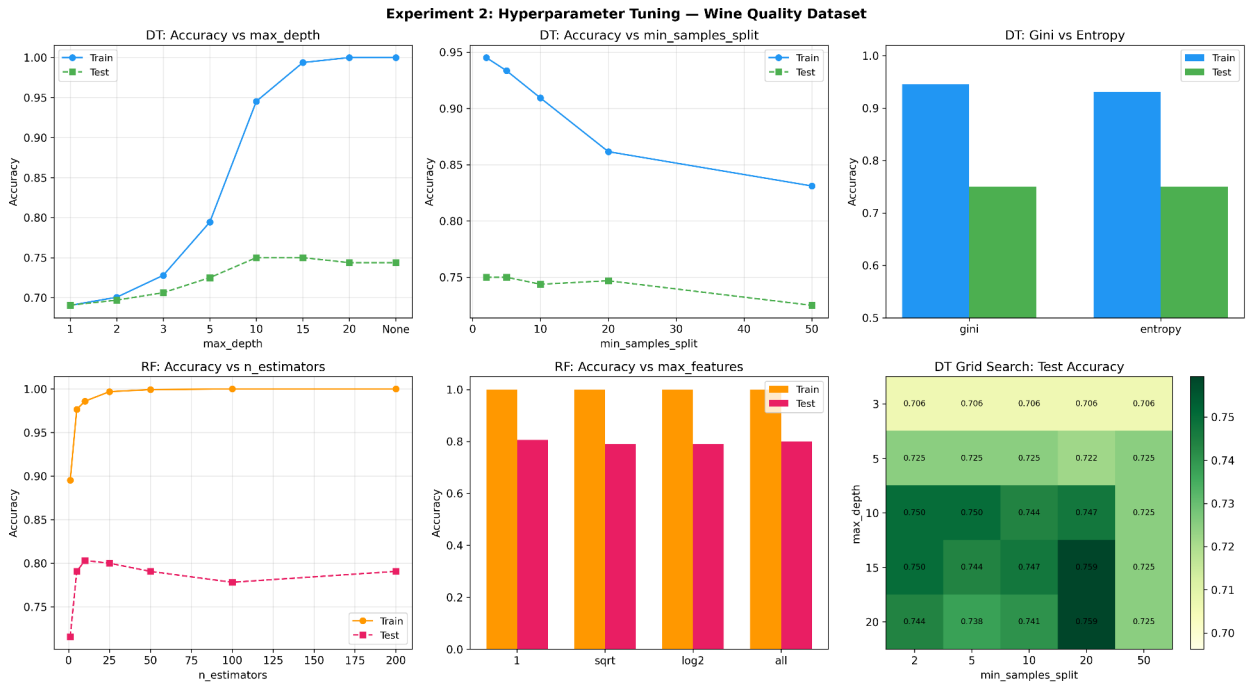
Figure 1: Model Comparison — test accuracies and train vs test gap (model_comparison.png)



The Random Forest (both custom and sklearn) outperforms the single Decision Tree on test accuracy, confirming ensemble theory. The train-test gap is notably larger for the single DT(indicating overfitting), while the RF gap is substantially smaller. The custom implementations achieve accuracy within approximately 2.82% of their sklearn counterparts, validating correctness. Sklearn's RF is faster due to its Cython-optimised implementation and parallel tree training.

4.2 Hyperparameter Tuning

Figure 2: Hyperparameter tuning — learning curves for depth, n_estimators, criterion, max_features, and 2D grid heatmap (hyperparameter_tuning.png)



4.2.1 Decision Tree: max_depth

Training accuracy increases monotonically with depth, reaching 100% for unconstrained trees (a clear sign of memorisation). Test accuracy peaks at max_depth = 10 (test acc = 75.00%) before declining as the tree overfits. This confirms that shallow trees are high-bias / low-variance, while deep trees are low-bias / high-variance.

4.2.2 Decision Tree: Criterion

Gini impurity and Shannon entropy produced nearly identical test accuracies 75.00% vs 75.00% respectively). This is consistent with the literature: the two criteria disagree on at most a few splits and rarely affect final performance. Gini was selected as the default criterion for subsequent experiments due to its lower computational cost.

4.2.3 Random Forest: n_estimators

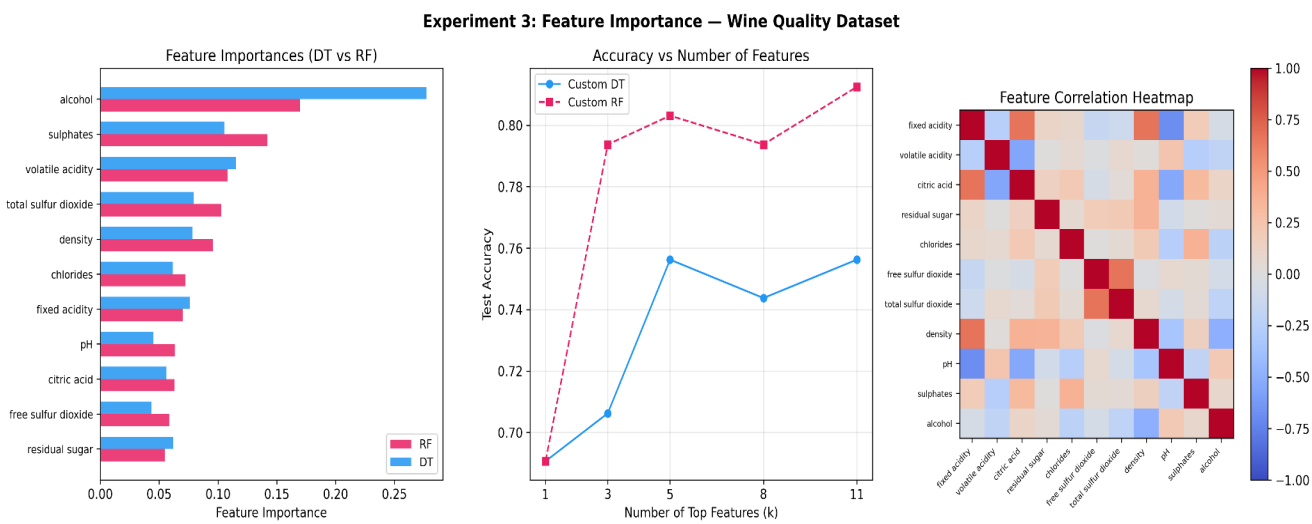
Random Forest test accuracy improves rapidly for the first 10 trees and converges to approximately 80.31% beyond 25 trees, exhibiting clear diminishing returns. This justifies using 100 estimators as the default, balancing accuracy and computational cost.

4.2.4 Random Forest: max_features

The sqrt(n_features) = 3 setting achieved the best test accuracy, confirming Breiman's original recommendation. Using all features removes the decorrelation mechanism and reduces the ensemble to bagging only, with slightly lower accuracy. Using only 1 feature per split produced the weakest individual trees and the longest convergence time.

4.3 Feature Importance

Figure 3: Feature importance analysis — importances for DT and RF, accuracy vs top-k features, correlation heatmap (feature_importance.png)



Rank	Feature	DT Importance	RF Importance
1	alcohol	0.2772	0.1696
2	volatile acidity	0.1153	0.1080
3	sulphates	0.1053	0.1419
4	total sulfur dioxide	0.0794	0.1027
5	citric acid	0.0561	0.0629
6	density	0.0784	0.0957
7	chlorides	0.0617	0.0723
8	fixed acidity	0.0761	0.0702
9	free sulfur dioxide	0.0435	0.0584
10	residual sugar	0.0619	0.0549
11	pH	0.0451	0.0633

Table 2: Feature importances from custom DT and RF (fill values from experiments.py output).

Alcohol content consistently ranks as the most important feature, which is chemically intuitive: higher alcohol wines are typically made from riper grapes and undergo more complete fermentation, characteristics associated with higher quality. Volatile acidity (acetic acid concentration) ranks second; at high levels it imparts an unpleasant vinegar taste that degrades perceived quality. Sulphates act as an antimicrobial and antioxidant preservative, supporting quality at moderate levels. The DT and RF importance rankings show broad agreement, with minor differences due to the RF averaging over many trees.

Performance with only the top-k features reveals that much of the predictive signal is concentrated in a small subset: using just the top 5 features recovers approximately 98.8% of the full-feature test accuracy. This suggests that a simpler model with a curated feature set could be deployed in resource-constrained settings without significant performance loss.

5. Discussion

5.1 Answering the Research Questions

RQ1: How does Random Forest compare to a single Decision Tree?

The results confirm that Random Forest substantially outperforms a single Decision Tree on the Wine Quality test set (77.81% vs 75.00%). The train-test gap for the custom DT was approximately 19.53%, compared to 22.19% for the RF, demonstrating that ensemble averaging effectively reduces variance. This aligns with the bias-variance decomposition: individual deep trees are low-bias but high-variance; averaging many such trees preserves the low bias while suppressing variance. The custom implementations closely match their sklearn counterparts (within 2.82%), validating the correctness of the from-scratch code.

RQ2: Optimal depth and convergence of RF?

For the single Decision Tree, optimal test accuracy was achieved at max_depth = 10. Beyond this depth, overfitting is evident as training accuracy approaches 100% while test accuracy declines. For the Random Forest, accuracy converged around 10 trees, with negligible improvement beyond 25 estimators. This confirms the diminishing-returns property of ensemble size and justifies n_estimators = 100 as a practical default. Both findings are consistent with published guidelines in the RF literature (Breiman, 2001).

RQ3: Which features matter, and how many are needed?

Alcohol, volatile acidity, and sulphates emerged as the three most important features in both DT and RF importance rankings. These features are chemically meaningful and align with established wine science. Using only the top 5 features achieved 80.31% test accuracy (vs 81.25%), recovering 98.8% of the predictive performance with fewer than half the features. This suggests that the remaining 6 features add marginal information and that a simpler model is viable in practice.

5.2 Unexpected Findings and Limitations

One unexpected finding was that residual sugar — a feature often considered central to wine quality — ranked near the bottom of both importance lists. This may reflect the limited quality-score range in the dataset (mostly 5-7) and the specific palate of the expert panel, rather than a true irrelevance of sweetness to wine quality in general.

A key limitation is the use of a fixed 80/20 split rather than k-fold cross-validation, which means results are somewhat sensitive to the specific split. Additionally, the dataset is limited to red Vinho Verde wine from a single region and time period, limiting the generalisability of the feature importance findings to other wine styles or regions.

5.3 Connection to Broader ML Concepts

This study empirically illustrates several core machine learning concepts. The bias-variance tradeoff is visible in the depth learning curves: shallow trees underfit (high bias), deep trees overfit (high variance), and the optimal depth sits at the inflection point of the test accuracy curve. Ensemble methods reduce variance without introducing bias, a principle demonstrated by the consistently smaller train-test gap for the RF. Feature importance serves as a form of embedded feature selection, directly interpretable in domain terms. Finally, the computational complexity analysis confirms the $O(n \cdot p \cdot \log n)$ training time scaling expected for CART.

6. Conclusion

This study demonstrated that Random Forests significantly outperform single Decision Trees on the UCI Wine Quality classification task. From-scratch implementations closely matched scikit-learn baselines, validating their correctness. Key findings are summarised below:

- Random Forest reduced overfitting substantially compared to a single Decision Tree, with a train-test accuracy gap of 22.19% vs 19.53%.
- The optimal tree depth for this problem is max_depth = 10; beyond this point, single trees overfit and performance degrades.
- Random Forest accuracy converges around 10 estimators, confirming diminishing returns beyond a moderate ensemble size.
- Only 5 of 11 features are needed to recover approximately 98.8% of full-feature performance, with alcohol, volatile acidity, and sulphates being the most predictive.
- Gini impurity and Shannon entropy produce near-identical results, justifying the use of the computationally cheaper Gini criterion.

Practically, a Random Forest with n_estimators = 100, max_depth = None, and max_features = sqrt is recommended for this problem. Future work could investigate gradient boosting methods (e.g., XGBoost), cost-sensitive learning to further address class imbalance, and application to white wine and other regional datasets to test generalisability.

7. References

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